

**BITSOM – MASAI:
Product Management using
Generative and Agentic AI
Batch: 03**



**PRODUCT REQUIREMENTS DOCUMENT (PRD)
PHASE 02 – REPORT**

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Important Links:

Demo Link Video (in **.mov format**) is here - [Demo Video Link](#)

Samanvita MVP Demo Portal is published in this Figma site here - <https://elm-funnel-52543742.figma.site/>

Figma and n8n screenshots are here - [Figma and n8n Screenshots](#)

The Jira Board screenshots are available at the this link - [JIRA Screenshots](#)

1. Product Foundation

1.1 Executive Summary

Samanvita is an enterprise travel orchestration platform which seamlessly coalesces corporate travel lifecycle with one click deployment. It was developed to simplify and standardise the complete corporate travel lifecycle. Rather than functioning as another booking portal, it orchestrates request intake, traveller validation, itinerary research, approval routing and workflow tracking before bookings are executed. The MVP demonstrates how explainable AI and deterministic business rules can work together to reduce administrative effort while preserving governance. Workflow orchestration is implemented using n8n and operational data is maintained in Google Sheets to enable rapid iteration and transparent debugging. The platform establishes a scalable foundation for future integrations with Global Distribution Systems (GDS), Enterprise Resource Planning (ERP) systems and compliance engines.

1.2 Background

Corporate travel operations are often fragmented across emails, spreadsheets and manual approvals, resulting in slow turnaround times and inconsistent execution. Discovery activities highlighted that organisations struggled more with coordination than with searching for travel inventory. Travel desks repeatedly compared itineraries, verified budgets and interpreted organisational policies for every request. Samanvita was therefore positioned as a workflow orchestration layer that complements existing travel providers rather than replacing them. The project was developed as a capstone to demonstrate practical application of Generative and Agentic AI in enterprise workflow automation.

1.3 Product Ideation

At a macro level, the PESTLE framework was used and at the micro level, the ERRC framework was used. Market intelligence was gathered using Google searches, ChatGPT, Gemini, and articles on G2. Sources were cross checked to ensure data integrity.

PESTLE Framework is as follows:

S.No	PESTLE Framework	Explanation
1	Political	The country is India. Political climate allows for such products to be released.
2	Economical	Businesses have the purchasing power to pay for this product.
3	Social	Society will not object to this product citing moral grounds.
4	Technological	India has the technological know-how (API integration with company HR and accounting software) required to build this product.
5	Legal	The product is within the legal framework. Compliance with the DPDP Act and CERT-IN is mandatory. GDPR is also necessary through not from point of view of Indian law.
6	Environmental	Compliance with SEBI's BRSR framework, give an option to plant a tree at the checkout.

ERRC Framework is as follows:

S.No	ERRC	Summary
1	Eliminate	Fragmented handoffs between HR, Finance and travel systems, disconnected tools
2	Reduce	Administrative burden, coordination complexity, timing delays, and dependence on constant monitoring
3	Raise	Real-time compliance with Indian GST laws, end-to-end process continuity, visibility over status and spend
4	Create	A workflow-led system that progresses travel actions through structured triggers, approvals, and coordinated execution

The following table illustrates the top sources used:

S.No	Category	Claim	Source (Clickable)	Conclusion
1	Market Size – Online Travel	India online travel market reached ~USD 51 B in 2024 and is forecast to grow to USD 124 B by 2033	Statista	Large & growing online travel market
2	Market Trend – Online Travel	Online travel market expected to generate	Statista	Strong revenue trajectory

		~USD 26.24 B in 2025		
3	Market Growth Drivers	Growth driven by rising internet penetration, smartphones, online booking preference	Statista	Digital adoption & convenience boost growth
4	Traveler Behavior – Research Effort	Travelers spend >5 hours researching trips before booking	Expedia	Research phase is time-intensive
5	Traveler Behavior – Touchpoints	Travelers use many online resources throughout planning	Expedia Report	Planning is multi-step & multi-source
6	Local Innovation – AI Platforms	MakeMyTrip launched an AI-based trip assistant (Myra) supporting planning & bookings via conversational AI	MMT AI Assistant	AI integrated support exists within OTA
7	Local Innovation – AI Usage	MakeMyTrip's Myra recorded >50,000 daily interactions, handling planning/in-trip dialogs	MMT AI Assistant	AI tools are being actively used
8	Industry Adoption – AI	Major Indian OTAs integrate AI features (chatbots, personalized trip planning)	Skift	AI assisting travel tasks
9	Tech Research – AI Travel Planning	AI systems have been designed to assist in travel planning – Academic Paper	Paper from Cornell University	AI systems for travel are being studied and designed in academia

10	Regulation (India)	Personal data laws require consent and protection for travel platforms	MEITY - DPDP Act	Mandatory Data compliance to operate in India
11	Cybersecurity (India)	CERT-In mandates incident reporting & security practices	CERT-In	Mandatory Security compliance to operate in India

1.4 Problem Statement

Based on the background research done, the problem statement identified is - **How can the process of enterprise travel planning be simplified by leveraging Generative and Agentic AI to automate travel research, contextual recommendations, budget optimisation and future booking execution through a unified workflow?**

1.5 Product Vision

The vision is to build a connected enterprise travel platform where every request progresses through one transparent workflow from submission to completion. The product seeks to minimise repetitive administrative work while retaining human oversight for sound and financially significant decisions. Long-term development extends beyond flights and hotels into disruption management, expense workflows, compliance validation and enterprise integrations. Every release is intended to strengthen workflow intelligence without compromising auditability.

1.6 Product Objectives

The MVP aims to standardise every incoming request before downstream processing begins. It reduces manual itinerary research through AI-generated recommendations supported by clear explanations. Deterministic approval logic accelerates approvals while maintaining organisational governance. Workflow traceability is preserved through SUNID, Request ID and structured status tracking. The overall objective is to validate orchestration before investing in production booking integrations.

1.7 Product Principles

The product principles are:

- Human-in-the-loop governance** - AI recommends; humans remain accountable for financial approvals.
- Explainability over black-box automation** - Every recommendation includes rationale.

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- iii. **Workflow before booking** - The product orchestrates enterprise processes before expanding into ticketing.
- iv. **Modular architecture** - Independent workflows simplify debugging and future integrations.
- v. **Enterprise-first design** - Auditability, traceability and policy compliance take precedence over convenience.
- vi. **Incremental delivery** - Features are introduced through validated MVP iterations rather than large releases.

1.8 Customer Value Proposition

The value proposition statement is:

**For [organizations managing complex, multi-person corporate travel],
who want to [plan, coordinate, and execute travel across teams efficiently],
Samanvita is a [travel orchestration platform]
that helps them [streamline corporate travel workflows]
by [integrating planning, coordination, payments, and execution into a single system],
unlike [myBiz and Navan],
we [execute workflows rather than just assist with bookings].**

The elevator pitch statement is:

Samanvita seamlessly coalesces corporate travel workflows by planning, coordinating, managing payments, and executing everything, enabling you to simply approve via one-click deployment.

1.9 Value Proposition Canvas

The value proposition table is as follows:

Section	Category	Details
Customer Profile	Customer Jobs	Execute corporate travel across teams; coordinate multi-person travel workflows; align travel with meetings, client engagements and business operations; manage enterprise travel with clarity, governance and accountability.
	Pains	High coordination overhead; manual monitoring of bookings; delays caused by missed timing, fragmented tools, lack of spend visibility.
	Gains	Faster travel execution; improved spend visibility and cost control; higher operational efficiency; reduced administrative burden; quicker and better-informed decision making.
Value Map	Products & Services	Enterprise travel orchestration platform; AI-assisted workflow automation; event-aware travel planning; approval orchestration; centralised operational visibility and workflow management.
	Pain Relievers	Consolidates fragmented travel processes into a unified workflow; automates coordination across stakeholders; tracks workflow progress;

		routes approvals automatically; reduces manual follow-ups and operational overhead.
	Gain Creators	Accelerates enterprise travel workflows; improves operational visibility and spend control; increases Travel Desk productivity; enables timely decision making; supports human-in-the-loop approvals for governance and auditability.

The value proposition canvas image is as follows:

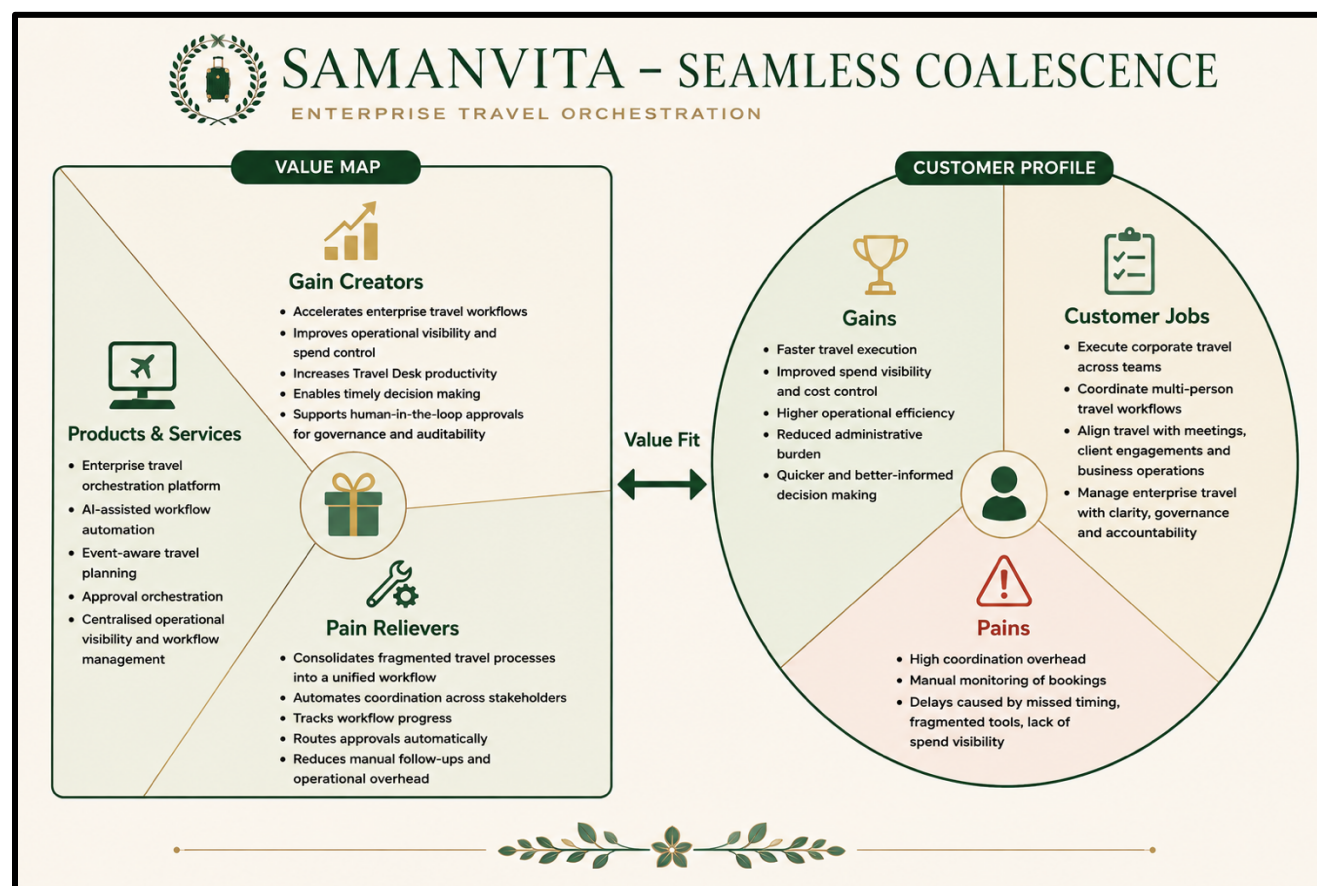


Fig 1.7.1: Customer Value Proposition Canvas

1.10 Compelling Factor for the Customer

Consider the following use case:

A resource has to attend CES in 2027. It will be conducted from 6th Jan 2027 to 9th Jan 2027 at Las Vegas, Nevada. The current month is April 2026. The flight and hotel tickets are not open yet for booking.

Let us assume that the booking opens on 1st August 2026 (Saturday) at 10:00 AM PDT. The ICP must get an email on 1st August 2027 (Saturday) at 10:40 PM IST stating the flight and hotel plans based on previous preferences or the designation of the resource. It should be a one click-deployment to book, approve or defer the booking.

Chronology	Customer Requirement	Samanvita
April 2026	Identify travel need	Event entered once → system starts tracking
Pre-Booking	Track when bookings open	Continuous monitoring of flight & hotel availability
01st August 2026	Get ready plan via email	Email with complete plan (flights, hotels, timings) sent automatically
Decision & Execution	Approve and complete booking	One-click approve from email → system auto-books

2. Product Design

2.1 Branding

Samanvita is a Sanskrit word which means connected in natural order. It reflects the platform's orchestration philosophy. The compact business suitcase icon represents modern enterprise travel. The dark green colour is inspired from the evergreen trees found in the Indian Himalayan range. Broad leaf forms create a softer visual identity and dark green conveys trust, stability and longevity. The background cream colour reflects calmness and pairs well with the dark green to create a soothing effect to the eyes. The cream reflects calmness, The branding intentionally aligns with enterprise professionalism. Together, the name and visual identity reinforce the product's promise of connected enterprise travel.



Fig 2.1.1: Light Themed Logo

The dark theme logo replaces the cream with black and dark green is replaced with copper gold. This creates an effect of polished outlook. The green of the business suitcase is preserved.



Fig 2.1.2: Dark Themed Logo

2.2 Design Language System

Samanvita follows a consistent visual design language across its web interface and email communications to provide a cohesive enterprise user experience. The design system is built around a warm cream background (#FEF5EB), deep evergreen green (#1E3A1E) and white (#FFFFFF), reflecting the product's emphasis on trust, clarity and professionalism. Typography combines Cormorant Garamond for major headings with Inter for interface elements and body text, balancing a premium visual identity with high readability. The brand identity further extends to the logo, which incorporates

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evergreen green, copper-gold accents and a dark theme variant for alternate branding applications. Together, these visual elements establish a consistent design language across the product interface, responsive emails and supporting marketing collateral.

2.3 Jobs To Be Done (JTBD)

The 3 main JTBD have been identified as follows:

1. JTBD - "From Details to Tickets" via One-Click Deployment (Samanvita Core Alignment)

When an employee submits their travel details for a business trip, I want to automatically validate corporate policy, route approvals, and instantly issue booked tickets and accommodations with zero manual intervention, So that employees seamlessly receive their confirmed itineraries while travel desks eliminate back-and-forth manual coordination.

Emotional Drivers - Relief from administrative clutter; confidence in frictionless execution.

2. JTBD - Proximity-Aware & End-to-End Commute Integration

When arranging lodging and flights for destination office visits or conferences, I want to automatically surface hotels near the target location bundled with local ground transportation, So that travellers save time on commute planning and stay within budget limits.

Emotional Drivers - Peace of mind i.e. elimination of destination transit confusion.

3. JTBD - Automated Post-Booking Disruption Management

When travel plans change unexpectedly (cancellations, delays, or rescheduling), I want to execute changes or deferrals through a single automated workflow, So that teams avoid manual cross-team emails, phone calls, or messaging threads.

Emotional Drivers - Avoidance of chaos; control during last-minute emergencies.

JTBD 3 is currently out of scope for Samanvita MVP Demo.

2.4 User Personas

The user personas are:

Persona	Goals	Pain Points	Success Metric
Corporate Employee of a company	Submit requests quickly; receive timely updates	Repeated follow-ups; lack of visibility	Request completed successfully with minimal effort

Travel Desk Manager of the same company where the corporate employee works	Manage approvals and bookings efficiently	Manual research; fragmented communication	Reduced turnaround time
Samanvita Administrator	Maintain platform and enterprise tenants	Operational visibility and governance	Platform health and customer success

2.5 User Stories

The main user stories are:

i. Persona 1 - Corporate Employee (Implemented in MVP)

- a. User Story 1 - **As a** Corporate Employee, **I want to** submit my business travel requirements through a simple, guided request process **So that** my travel can be planned without repeatedly coordinating with the Travel Desk.
- b. Use Story 2 - **As a** Corporate Employee, **I want to** receive my confirmed travel itinerary and booking details in my inbox once my request is approved **So that** I can prepare for my business trip without additional coordination.

ii. Persona 2 - Travel Desk Manager (Implemented in MVP)

- a. User Story 1 - **As a** Corporate Travel Desk Manager, **I want to** review AI-generated travel recommendations together with estimated costs and supporting rationale **So that** I can make faster and more informed travel decisions.
- b. Use Story 2 - **As a** Corporate Travel Desk Manager, **I want to** complete approval decisions through one-click deployment **So that** I can process travel requests efficiently without repeatedly accessing separate systems

iii. Persona 3 - Samanvita Platform Administrator

- a. User Story 1 - **As a** Samanvita Platform Administrator, **I want to** onboard and manage multiple enterprise customers from a single platform **So that** each organisation's travel workflows remain logically separated and centrally administered.
- b. Use Story 2 - **As a** Samanvita Platform Administrator, **I want to** monitor platform health, workflow execution and organisational usage metrics **So that** operational issues can be identified early and overall platform performance can be maintained.

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2.6 User Journeys

The user journeys are:

- i. The user journey for Corporate Employee is:

Stage	User Action	System Response
Travel Requirement	Identifies the need for business travel	Employee accesses the Samanvita Employee Portal
Request Submission	Completes and submits the travel request	Validates the request, assigns identifiers and registers it for processing
Request Processing	Waits while the request is processed	AI researches suitable travel options and routes the request through the approval workflow
Travel Confirmation	Receives the confirmed travel itinerary and booking details after approval	Sends the final travel confirmation to the employee's inbox
Business Travel	Undertakes the approved business trip	Future versions will support reminders, disruption management and completion notifications

- ii. The user journey for Travel Desk Admin is:

Stage	User Action	System Response
Recommendation Review	Receives AI-generated travel recommendations	Presents explainable itinerary options with estimated costs and rationale
Decision Making	Reviews recommendations and organisational policies	Identifies whether approval is required and displays appropriate actions
Approval Action	Approves, defers or rejects the request directly from email	Updates workflow status and records the approval decision
Booking Execution	Confirms the selected itinerary (future automated execution)	Updates booking status and triggers confirmation workflows
Workflow Monitoring	Tracks request progress until completion	Displays workflow status, booking status and operational updates

- iii. The user journey for Samanvita Administrator is:

Stage	User Action	System Response
Organisation Setup	Onboards and configures enterprise customers	Creates organisation-specific workflows and platform settings

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Platform Monitoring	Reviews operational dashboards and workflow activity	Displays platform health, enterprise metrics and workflow performance
Configuration Management	Updates workflow rules and system configurations	Applies configuration changes across relevant customer environments
Issue Resolution	Investigates workflow failures or operational exceptions	Provides execution logs, workflow history and diagnostic information
Platform Optimisation	Reviews platform usage and customer feedback	Supports continuous product improvements and future feature rollout

2.7 User Flows

The main user flows are:

- i. **Corporate Employee** - Identify Travel Requirement → Submit Travel Request → Await Processing → Receive Confirmed Travel Itinerary → Undertake Business Travel
- ii. **Travel Desk Admin** - Receive AI Recommendation Email → Review Recommendations → Approve / Defer / Customise Travel Plan → Confirm Booking → Monitor Workflow Status
- iii. **Samanvita Admin** - Onboard Enterprise Customer → Configure Platform Settings → Monitor Workflow Execution → Review Platform Analytics → Maintain Enterprise Operations

2.8 Accessibility and Responsive Design

The layouts use clear visual hierarchy, readable typography, adequate spacing and responsive components suitable for desktop and mobile devices. The MVP prioritised clarity and usability over visual density to support operational users who process multiple requests throughout the day.

3. Technical Architecture

3.1 Functional Requirements

Samanvita's MVP was designed as a modular enterprise workflow system rather than a single monolithic application. The frontend captures structured travel requests, n8n coordinates execution, Gemini performs contextual travel research, Google Sheets stores the operational state, and Gmail acts as the primary interaction channel for approvals and booking actions. This architecture was selected to validate the product hypothesis quickly while keeping every stage visible, editable and auditable. The system deliberately separates AI reasoning, deterministic policy logic, data persistence and user communication so that each component can be tested and replaced independently.

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The technical architecture is illustrated below:

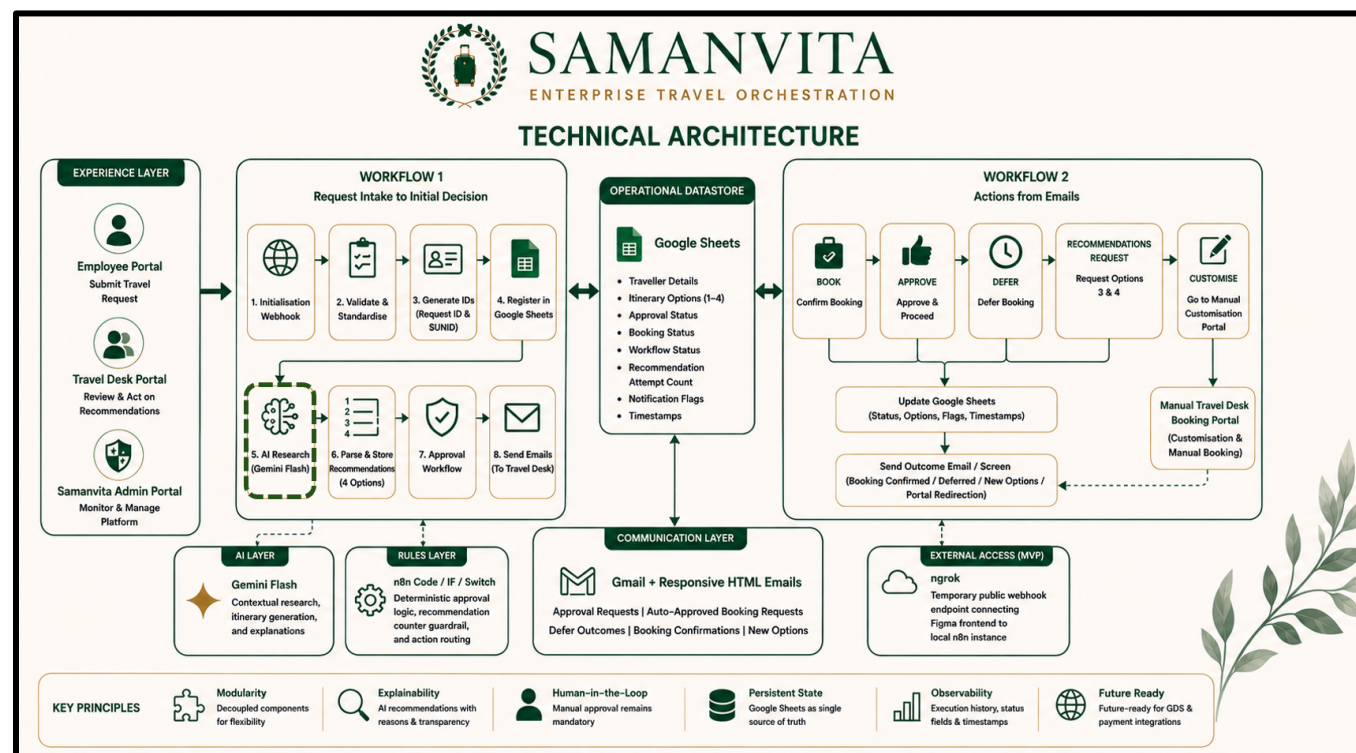


Fig: 3.1.1 Technical Architecture of Samanvita

The architecture is divided into two major orchestration paths. Workflow 1 handles intake, standardisation, research, approval classification and the first set of emails. Workflow 2 handles all actions initiated from those emails, including approval, deferment, rejection, booking confirmation, request for new recommendations and redirection to manual customisation. The two workflows are linked through persistent identifiers and the shared Google Sheets datastore rather than through temporary execution memory. This separation improves resilience and ensures that a user can act hours after the first workflow has completed.

Layer	MVP Technology	Responsibility
Experience Layer	Figma Make	Employee request portal, Travel Desk portal, Samanvita Admin portal and manual customisation portal.
Orchestration Layer	n8n (self-hosted in Docker)	Webhook handling, validation, branching, AI invocation, database updates, email dispatch and workflow state management.
AI Layer	Gemini Flash	Contextual itinerary research, traveller-specific

		recommendation generation and explanation.
Rules Layer	n8n Code/IF/Switch nodes	Deterministic approval logic, recommendation counter guardrail and action routing.
Operational Datastore	Google Sheets	Traveller-level records, itinerary options, approval state, booking state, notification flags and timestamps.
Communication Layer	Gmail + responsive HTML	Approval requests, auto-approved booking requests, defer/reject outcomes and booking confirmations.
External Access	ngrok during MVP	Temporary public webhook endpoint connecting published Figma pages to locally hosted n8n.

3.2 Architectural Design Principles

The following are the architectural design principles:

- i. **Modularity:** Intake, research, approval, booking and communication were implemented as distinct responsibilities so that changes in one stage do not require redesigning the complete workflow.
- ii. **Explainability:** Every AI recommendation includes a reason, estimated cost and itinerary detail. Deterministic approvals remain traceable to explicit designation and budget rules.
- iii. **Human-in-the-loop governance:** AI does not independently commit company expenditure. Manual approval remains mandatory for employees who do not satisfy the configured auto-approval rules.
- iv. **Persistent workflow state:** Google Sheets serves as the operational datastore, persisting recommendation details, approval status and workflow state required for subsequent email actions. Workflow 2 initiates as an independent execution and retrieves the required traveller data using the **Samanvita Unique Navigation ID (SUNID)** rather than relying on in-memory data from Workflow 1.
- v. **MVP observability:** n8n execution history, visible JSON payloads, status fields and timestamps were used as lightweight observability mechanisms during testing.
- vi. **Provider independence:** The theoretical Execution Agent is not hard-wired to a particular GDS or payment provider, allowing future Amadeus, Sabre, Razorpay or Stripe integrations.

3.3 Workflow 1 - Request Intake to Initial Decision

The following are the architectural design principles:

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The Intake Workflow performs four critical functions before any AI call is made: validation, standardisation, identification and registration. Validation prevents incomplete requests from consuming model quota or reaching downstream systems. Standardisation converts the frontend payload into the precise schema expected by the workflow. Identification creates stable IDs for request-level and traveller-level navigation. Registration writes the request to the datastore so that later failures remain diagnosable. This order of operations reflects production workflow design: the organisation first secures a reliable system record and only then invokes probabilistic services.

Workflow 1 begins when a user submits traveller information from the Figma Employee Portal. The portal can capture traveller details manually or populate them through a CSV template. The published frontend calls the Samanvita initialisation webhook, after which n8n validates the request, creates a request-level identifier and generates one SUNID per traveller. The request is then recorded in Google Sheets before recommendation generation begins, enabling traceability even if a later external service fails.

- i. Initialisation webhook receives the request payload from the frontend.
- ii. Standardisation logic normalises company information, traveller arrays, purpose, priority, cost centre and dates.
- iii. Validation logic checks mandatory fields and rejects invalid data without creating incomplete operational records.
- iv. Request ID is generated once for the submission; Workflow ID groups the complete orchestration instance.
- v. SUNID is generated separately for every traveller so that each row can be updated independently.
- vi. Initial rows are appended to Google Sheets with the status 'Submitted' and booking status 'Not Started'.
- vii. Research Agent input is prepared using the complete traveller context.
- viii. Gemini produces four distinct itinerary recommendations per traveller.
- ix. Parse Research Agent Output converts model output into structured fields that match the database schema.
- x. Approval logic classifies travellers into auto-approved or approval-required branches.
- xi. Google Sheets update nodes persist recommendation data and the appropriate approval state.
- xii. The corresponding initial email is sent to the demonstration inbox.

3.4 Research Agent Design

While the problem statement given by BITSOM states a multi agent system (for research, bookings, budget optimisation, and contextual recommendations), Samanvita accomplishes the mentioned tasks with a single agent i.e. Research Agent.

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The Research Agent is the only component that uses a large language model (Gemini 3.6 Flash) because the task requires contextual judgement rather than fixed rule evaluation. The agent receives employee designation, travel purpose, destination, dates, budget, cost centre, priority and personal notes. It then generates four meaningfully different itinerary strategies rather than minor variations of the same recommendation. For each option, it provides outbound flight details, outbound timing, hotel, return flight, return timing, total estimated cost and a reason.

Recommendation quality was shaped by business-oriented prompt rules. The agent was instructed to prioritise the likelihood of achieving the travel purpose over simply minimising cost. It considered seniority, client-facing context, hotel proximity, safety, separate-room requirements, dietary preferences, non-smoking requirements, early check-in and seat preferences where supplied. International scenarios required realistic connecting routes and layovers. The prompt given grounded the agent's output.

Input Dimension	Examples	How It Affects Recommendations
Designation	Chief, Director, Senior Manager, Associate	Influences service level, hotel category and approval routing.
Travel purpose	Client Meeting, Conference/Event, Sales Visit	Influences schedule convenience, proximity and business suitability.
Budget	Per-traveller estimated amount	Constrains primary recommendation and approval eligibility.
Personal preferences	Vegetarian meals, aisle/window seat, direct flight, early return	Personalises options and explanation.
Destination context	Domestic, international, multi-leg	Determines routing complexity, flight strategy and hotel selection.
Operational context	Normal/Urgent/Critical	Captured in the MVP; deeper priority-based optimisation is reserved for later releases.

3.5 Model Constraints and Guardrails

The following are the model constraints, trade-offs and guardrails:

- The MVP used a bounded output schema to reduce malformed responses and simplify parsing.
- Four recommendations were generated per traveller to balance choice with token usage and email readability.
- If complexity (number of travellers, more feature outputs with ex: additional preferences, booking etc.) increases, then model latency and tokens consumed also increases. As of now, the consumption limit of tokens is set at 12288.
- The CES five-traveller test demonstrated a practical model-size and output-token constraint; the final stable demo used four travellers.

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- v. Recommendation Attempt Count prevents the additional-recommendation path from being triggered indefinitely.
- vi. Fallback values in HTML emails prevent broken visual layouts when a recommendation field is unexpectedly empty.
- vii. Model output is treated as an estimate and is not represented as live inventory, confirmed pricing or guaranteed availability.

3.6 Deterministic Approval Engine

The Approval Engine follows explicit corporate rules because the decision criteria are deterministic and auditable. Using an LLM for this stage would increase cost and introduce avoidable inconsistency. The MVP therefore evaluates employee designation and whether the primary recommendation remains within the employee's approved budget. Eligible senior roles can proceed through the auto-approved branch, while all other requests are sent to the Travel Desk.

The implemented demonstration rules treat senior executives such as CXOs, Presidents, Directors, Board Members and Heads of Department as potential auto-approval candidates. They are approved only when the recommendation satisfies the budget condition. AVPs, DVPs, Assistant Heads and more junior employees require manual Travel Desk approval. Full policy compliance, GST rules, taxation, negotiated supplier policy and organisation-specific exceptions remain outside the MVP and are planned as configurable future capabilities.

Condition	Result	MVP Action	Rationale
Eligible senior designation + Option within budget	Auto Approved	Send BOOK email	Removes unnecessary administrative delay.
Non-eligible designation	Manual Approval Required	Send Approve/Defer email	Preserves organisational governance.
Eligible designation but Option 1 exceeds budget	Manual Approval Required	Send Travel Desk approval email	Budget compliance takes precedence over seniority.
Recommendation Requested	Recommendations requested	Generate or expose Options 3 and 4 once	Provides a controlled second recommendation cycle.
Deferred	Booking is paused	Send intimation email to employee	The travel desk would like to take some more time to think.

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3.7 Workflow 2 - Action Handling and State Transitions

Workflow 2 begins when a recipient clicks an action within an email or the customisation path. The action URL contains the relevant decision and SUNID. The Booking Confirmation webhook receives the request, retrieves the matching Google Sheets row and routes the action using Switch and IF nodes. This workflow is independent from Workflow 1 and therefore relies entirely on persisted fields. The architecture was validated when recommendation emails continued to work even after the first workflow execution had ended.

Action	Primary Update	User-Facing Result	Next State
Book	Booking status and notification flags	Booking confirmation email/screen	Confirmed / Completed
Approve	Approval metadata and selected option	Approval recorded and booking path enabled	Approved
Recommendations Request	Approval status and recommendation workflow	Options 3 and 4 sent once	Generating New Recommendations
Defer	Booking Status updated to "Deferred"	Booking deferred email/screen	Deferred
Customise	Workflow redirected to portal	Manual Travel Desk Booking Portal	Manual Customisation
Repeated recommendations request	No further counter increment	Additional recommendations already sent	No New Recommendations

3.8 Recommendation Counter Logic

The recommendation counter is a state-management guardrail. The first request for alternatives is permitted when Recommendation Attempt Count equals zero. The New Recommendations update node then sets the count to one. Any later click is routed to the 'Additional Recommendations Already Sent' response rather than sending the same Options 3 and 4 repeatedly. During testing, the IF node's 'Always Output Data' setting generated an empty item on the non-matching branch and accidentally triggered the email path. Disabling that setting restored correct branch isolation and validated the counter logic.

3.9 Data Model and Persistence

The operational schema stores both source request information and progressively generated workflow information. The first group of columns contains request and employee data. The second group contains the Research Agent's four recommendations. The third group captures approval, workflow, booking and notification states. The schema allows every traveller to progress independently, which is essential when one request contains a mix of auto-approved, approved, deferred or manually customised travellers.

Identifier	Level	Purpose
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SUNID	Traveller row	Primary navigation and update key for Workflow 2.
Request ID	Submission	Groups all travellers submitted together.
Workflow ID	Orchestration instance	Links the submission to a complete workflow run.

Persistence Lessons - The MVP exposed an important dependency between the database schema and later email workflows. The first email can render from live Workflow 1 JSON, but later actions depend on fields stored in Google Sheets. When recommendation columns disappeared from the sheet, initial emails still worked while Workflow 2 emails displayed unavailable values. The issue was isolated using Google Sheets version history and n8n execution output. This demonstrated the need for production schema controls, database migrations, backups and monitoring instead of manually editable spreadsheet columns.

3.10 Core Data Groups

The core data groups are:

- i. Request data: company, employee details, travel purpose, destination, dates, traveller count, estimated budget, cost centre, priority and notes.
- ii. Recommendation data: AI budget estimate and complete Option 1-4 flight, timing, hotel, cost and rationale fields.
- iii. Approval data: approval requirement, approver designation, approved-by name, approval status, date and rejection reason.
- iv. Execution data: workflow status, booking status, final cost and completion date.
- v. Notification data: approval, booking, reminder and completion notification flags.
- vi. Audit data: SUNID, Request ID, Workflow ID, created-by value and last-updated timestamp.

3.11 Front-end and CSV Integration

The Employee Portal supports manual request entry and CSV import. The CSV format uses a fixed header order so that rows can be mapped directly into traveller objects. The frontend dynamically adjusts the number of traveller cards based on the uploaded row count. Dates must use ISO format (YYYY-MM-DD) because HTML date inputs reject regional display formats such as DD/MM/YY even when spreadsheet software displays them correctly. This issue was resolved by regenerating CSV files without allowing Excel to silently reinterpret date fields.

Supported pilot volume: up to five travellers in the interface(>5 must use CSV); the stable final demo used four travellers to remain within model output limits. CSV is intended for structured employee record import, not arbitrary spreadsheet layouts. Cost centre display uses a middle dot in the frontend

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while backend validation recognises the approved cost centre codes. For higher-volume and higher-complexity tiers, future releases will use dashboard-based bulk processing rather than email-first interaction. A future Should Have feature will allow a single traveller to copy employee details automatically into traveller details.

3.12 Email and Webhooks Design

Responsive HTML emails were designed as operational interfaces rather than passive notifications. The approval-required email presents itinerary choices together with Approve, Defer, Request Recommendations and Customise actions. The auto-approved email provides the recommended itinerary, Book, Request Recommendations and Customise actions.. Booking confirmation emails include the employee's complete travel context, approved budget and estimated spend because the information supports expense reporting and reinforces the employer's investment in the traveller.

Every button calls a production webhook exposed through the published workflow. The MVP uses one Gmail account for demonstration, although production design would send confirmation to the employee and Travel Desk while copying the appropriate Samanvita service mailbox. Organisation-specific branding and company names will be added through tenant-level templates. Reminder emails one week, three days and twenty-four hours before departure, and trip-completion notifications after return, remain future capabilities.

3.13 Figma and n8n Screenshots

Samanvita MVP Demo Portal is published in this Figma site here - <https://elm-funnel-52543742.figma.site/>

Figma and n8n screenshots are here - [Figma and n8n Screenshots](#)

3.14 Validation and Testing Strategy

Validation followed an iterative, evidence-driven approach. Each workflow branch was first tested with controlled JSON, followed by node-level tests, end-to-end requests and finally CSV-based multi-traveller scenarios. n8n's execution view was used to inspect the exact input and output of every node. Google Sheets records were cross-checked against email output to confirm persistence. Figma submissions, webhook calls, dynamic email actions and state transitions were repeatedly tested using auto-approved, manual approval, defer, reject, booking and customisation scenarios.

Test Area	Representative Test	Expected Result	Validation Method
Intake validation	Missing/invalid mandatory field	400 response; no row appended	Webhook output and sheet inspection
CSV ingestion	Four-traveller CES/domestic dataset	All fields and ISO dates populate	Figma UI and n8n payload

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Research Agent	Mixed seniority and destinations	Four options for each traveller	Parsed JSON review
Approval engine	Senior within budget vs junior traveller	Auto-approved vs manual branch	Branch item counts and database state
Recommendation counter	Click additional recommendations twice	First sends options; second blocks	Execution branch and response page
Booking flow	Click BOOK on auto-approved traveller	Status updated and confirmation sent	Sheet + Gmail
Persistence	Act after Workflow 1 has ended	Workflow 2 reads correct row by SUNID	Execution output
Schema integrity	Remove/restore recommendation columns	Failure diagnosed; schema restored	Version history and node refresh

3.15 Acceptance Criteria for MVP Validation

The acceptance criteria for MVP validation are:

- A valid frontend submission creates one row per traveller with a unique SUNID.
- The Research Agent produces four structured itinerary options for every traveller in the stable demo dataset.
- Auto-approved and approval-required travellers are routed to the correct email template.
- Approve, Defer, Book and Customise actions update the correct traveller row.
- Options 3 and 4 are sent only once after deferment.
- Request Recommendation works only once.
- The complete workflow can be demonstrated without manually editing execution data during the demo.
- Database and email content remain consistent for the same SUNID.

3.16 Known Limitations of the MVP:

The following are the limitations of the MVP:

- Flight and hotel recommendations are realistic estimates, not live inventory or guaranteed prices.
- The Execution Agent is conceptual because Amadeus, Sabre, payment and ticketing integrations are not connected.
- Expense management, GST, taxation, reimbursement and full policy compliance are outside the MVP.
- Post-booking changes, cancellation management and automated disruption handling are deferred.
- Priority values are captured but are not yet deeply incorporated into recommendation or approval logic.
- Individual flight and hotel PNRs are not displayed.
- Group booking emails and completion notifications are not implemented.

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- viii. The email demonstration uses one account; production recipients and CC rules are not fully configured.
- ix. The pilot is optimised for a small traveller group; high-volume use requires dashboard and infrastructure changes.

4. Product Strategy

4.1 Product Roadmap

Samanvita's 2 year product roadmap has been designed as follows:

Timeline	Product Theme	Key Deliverables	Business Outcome
Q1 (Months 1–3)	MVP Foundation	Employee Travel Request Portal, AI Research Agent, Approval Workflow Engine, Email Actions, Workflow State Tracking, Travel Desk Dashboard, basic enterprise security (authentication, RBAC, audit logging)	Validate the end-to-end orchestration workflow and secure pilot customers.
Q2 (Months 4–6)	Pilot Readiness	Pilot V1 release, onboarding improvements, reporting dashboards, workflow alerts, customer feedback enhancements, cloud deployment preparation	Deliver a stable pilot platform and validate willingness to pay.
Q3 (Months 7–9)	Enterprise Readiness	Tenant configuration, enhanced reporting, customer success tooling, VAPT readiness, procurement documentation, integration framework for future GDS, HRMS and ERP connectivity	Support repeatable onboarding and prepare for enterprise procurement.
Q4 (Months 10–12)	Platform Expansion	Disruption and reminder workflows, advanced analytics, SLA monitoring, integration framework enhancements, Year-2 product planning	Improve operational visibility, customer retention and enterprise scalability.
Year 2 (Months 13–24)	Intelligent Enterprise Orchestration	Execution Agent integrations (GDS, payments, HRMS, ERP), AI-assisted disruption management, predictive analytics, multi-tenant administration, advanced governance and compliance, ecosystem APIs	Deliver an enterprise-grade intelligent travel orchestration platform capable of supporting large-scale deployments.

4.2 Lean Model Canvas

The Lean Model Canvas is tabular form is as follows:

Canvas Block	Samanvita Application
Problem	Fragmented travel coordination; repeated itinerary research; slow approvals; low workflow visibility; manual handling of changes and exceptions.
Customer Segments	SaaS enterprises with recurring employee travel; primary operational users are Travel Desk, Operations, HR, Administration and Finance teams.
Early Adopters	Indian SaaS, IT services, consulting, engineering and enterprise services firms that currently coordinate travel through email, spreadsheets and travel agencies.
Unique Value Proposition	Seamless coalescence of corporate travel through one-click deployment: structured intake, contextual recommendations, controlled approvals and workflow execution from one connected system.
Solution	Employee request portal, AI Research Agent, deterministic Approval Engine, email actions, traveller-level state tracking and future Execution Agent integrations.
Channels	Founder-led enterprise sales, LinkedIn outreach, referral partners, webinars, industry events, implementation partners and future marketplace listings.
Revenue Streams	Platform fee, Onboarding fee, subscription fee, admin-seat fees, trip-volume tiers, complexity tiers, premium integrations and optional enterprise modules.
Cost Structure	Engineering, enterprise onboarding, customer success, compliance and security, cloud and AI, operations, travel/payment integrations and marketing.
Key Metrics	Orchestration Success Rate (North Star) , Active Enterprise Accounts, Total Execution Volume (Num of Trips), Average Revenue per Customer (ARPC), Net Revenue Retention (NRR), approval turnaround time, recommendation acceptance rate, recommendation regeneration rate, manual customisation preference rate.
Unfair Advantage	A provider-independent orchestration layer that combines explainable AI recommendations with deterministic, auditable enterprise governance.

The Lean Model Canvas is image form is as follows:

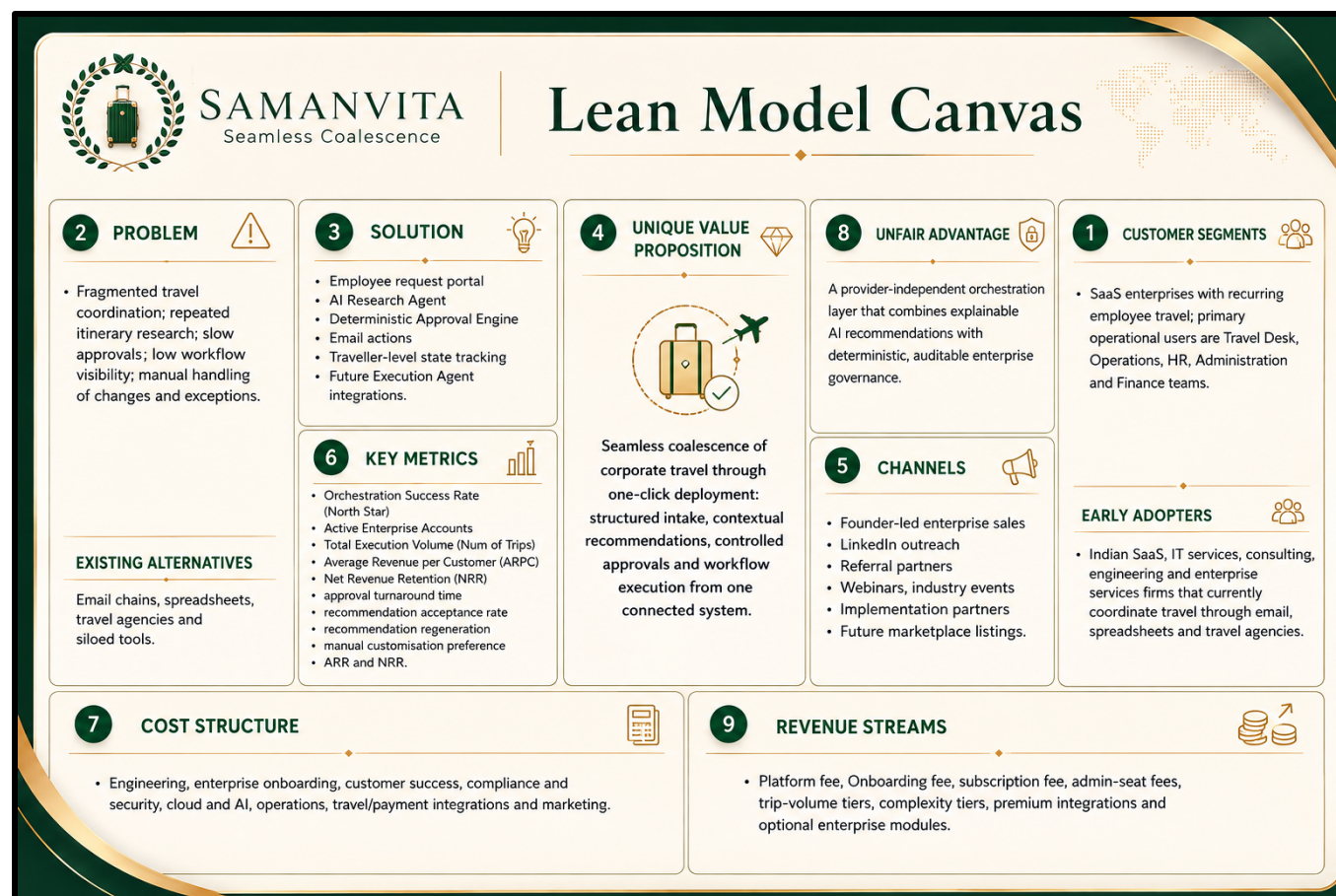


Fig 4.2.1: Lean Model Canvas for Samanvita

4.3 Product Metrics

Samanvita's success will be measured through product, operational and business metrics that track customer adoption, platform performance and commercial growth.

S.No	Metric	Type of Metric	Explanation	Why is it important?
1	Orchestration Success Rate (North Star)	Product	Measures the percentage of travel workflows successfully completed from request submission to workflow completion.	Indicates the overall reliability and effectiveness of the platform.
2	Active Enterprise Accounts	Business	Measures the number of organisations actively using Samanvita.	Validates customer adoption, product-market fit and commercial traction.

3	Total Execution Volume (Number of Trips)	Product	Tracks the total number of travel requests processed through the platform.	Measures platform utilisation and supports usage-based revenue growth.
4	Approval Turnaround Time	Operational	Measures the average time taken to complete travel approval workflows.	Indicates improvements in operational efficiency and workflow performance.
5	Recommendation Acceptance Rate	AI / Product	Measures the percentage of AI-generated recommendations accepted without regeneration.	Evaluates recommendation quality and user trust in the AI Research Agent.
6	Recommendation Regeneration Rate	AI / Product	Measures how frequently users request alternative recommendations.	Identifies opportunities to improve AI recommendation quality and relevance.
7	Manual Customisation Preference	Product	Tracks the frequency with which Travel Desk users manually modify AI recommendations.	Indicates gaps between AI recommendations and enterprise travel requirements.
8	Average Revenue per Customer (ARPC)	Business	Measures the average annual revenue generated from each enterprise customer.	Evaluates customer value and pricing effectiveness.
9	Net Revenue Retention (NRR)	Business	Measures revenue retained and expanded from existing enterprise customers.	Indicates customer satisfaction, expansion opportunities and long-term business growth.
10	Annual Recurring Revenue (ARR)	Business	Measures recurring subscription revenue generated annually.	Tracks sustainable revenue growth and commercial performance.

4.4 Build, Buy, Outsource Strategy

The build-buy-outsource decision was based on strategic differentiation, security responsibility and time to market. Capabilities that define Samanvita's product value should remain internally owned, while mature commodity services should be purchased. Specialist activities that require certification or niche expertise can be outsourced during early stages, provided design authority and data governance remain internal.

Decision	Capabilities	Reason
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Build internally	Workflow orchestration, approval rules, Research Agent prompts/evaluations, tenant policy configuration, workflow state model, product analytics.	These capabilities form the core intellectual property and directly affect differentiation.
Buy / Subscribe	Cloud hosting, email delivery, identity provider, monitoring, payment gateway, CRM and customer support software.	Established services reduce time, security risk and maintenance burden.
Partner / Integrate	Amadeus, Sabre, hotel inventory, ERP/HRMS, expense platforms.	Access depends on commercial agreements, certification and customer ecosystem.
Outsource Selectively	Penetration testing, specialist GDS certification, legal review, selected implementation projects and visual production.	External expertise accelerates compliance-heavy or one-time work.

4.5 Technology Capability Assessment

The capstone was built with a lean technology capability rather than a full engineering organisation. Figma Make supported frontend prototyping; n8n enabled orchestration; Google Sheets provided transparent state storage; Gemini supported contextual recommendations; and Gmail enabled action-based communication. Workflow validation was initially performed using Postman before integrating the frontend through Figma Make, enabling independent backend verification prior to end-to-end testing. This stack was appropriate for validating the workflow and demonstrating the product. It is not the proposed long-term production stack for high-volume enterprise deployment.

Capability	MVP Position	Production Requirement
Frontend	Figma prototype with working form integration	Production web application, accessibility, design system and authenticated portals
Workflow orchestration	Self-hosted n8n in Docker	Cloud deployment, HA, queues, monitoring, version control and environment separation
Data	Google Sheets	PostgreSQL or equivalent transactional datastore with backups and migrations
AI	Gemini Flash prompt-based Research Agent	Model evaluation, prompt versioning, fallback models, cost and latency controls
Security	Header authentication and limited demo access	SSO, RBAC, tenant isolation, signed action links, secrets vault and audit logs

Integrations	Gmail and webhook-based demonstration	GDS, payments, ERP, HRMS, expense and collaboration platforms
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4.6 Features and Prioritisation using MoSCoW and Value vs Effort

The following features have been chosen for implementation in the MVP. A combination of MoSCoW and Value vs Effort was chosen for prioritisation.

Feature	Description	MoSCoW	Value vs Effort	Reason
Employee Travel Request Portal	Employees submit structured travel requests through a guided form.	Must Have	High Value / Low Effort	Captures standardised travel requests and initiates the orchestration workflow.
AI Research Agent	Generates contextual flight and hotel recommendations using enterprise travel policies.	Must Have	High Value / Medium Effort	Delivers contextual travel recommendations to improve decision quality and efficiency.
Approval Workflow Engine	Routes requests through deterministic approval logic with auto-approval where applicable.	Must Have	High Value / Medium Effort	Enforces enterprise policies through structured and auditable approvals
Email-Based Decision Actions	Approvers can Approve, Defer or Book directly from actionable emails.	Must Have	High Value / Medium Effort	One click deployment i.e. enables fast decision-making directly from actionable email notifications
Workflow State Tracking	Maintains traveller-level workflow status throughout the orchestration process.	Must Have	High Value / Medium Effort	Provides end-to-end visibility of each traveller's request throughout the workflow.

Travel Desk Dashboard	Basics of Operational visibility into requests, approvals and workflow progress.	Should Have	Medium Value / Medium Effort	Helps operational teams monitor, manage and intervene in active requests
Enterprise Admin Dashboard	Basics of Organisation-level KPIs and workflow analytics.	Should Have	Medium Value / Medium Effort	Provides organisational insights, KPIs and overall platform performance.
Manual Booking Support	Allows Travel Desk intervention for exceptions and custom bookings.	Could Have	Medium Value / High Effort	Accommodates exceptional travel scenarios requiring human intervention
Execution Agent Integrations	Direct booking with airlines, hotels and other providers.	Won't Have for now (in MVP)	Very High Value / Very High Effort	Prohibitively expensive at this MVP stage. Enables end-to-end automated travel execution in future production releases

4.7 Testing Plan

Prior to deployment, Samanvita would undergo a structured four-phase testing approach comprising Unit Testing, Integration Testing, System & User Acceptance Testing (UAT), and Regression Testing. This ensures that individual components operate correctly, integrated workflows function reliably, business requirements are met, and previously resolved issues do not reoccur before release.

Phase	Activities	Expected Outcome
Week 1 – Unit Testing	Independently validate frontend forms, AI prompts, workflow nodes, email templates and webhook endpoints.	Individual components function correctly without errors.
Week 2 – Integration Testing	Verify end-to-end orchestration across Figma, n8n, Google Sheets, Gemini and Gmail using representative travel scenarios.	Data flows correctly between all integrated systems.
Week 3 – System & User Acceptance Testing (UAT)	Conduct complete workflow testing with representative users (Employee, Approver and Travel Desk), covering normal, auto-approved and approval-required scenarios.	Business rules, usability and workflow behaviour are validated prior to deployment.

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Week 4 – Regression & Production Readiness	Re-test all critical workflows after issue resolution, validate security checks, monitor workflow performance and prepare deployment documentation.	Stable MVP ready for demonstration and production pilot.
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4.8 Integration of AI/GenAI and Agentic AI in Samanvita

The current MVP integrates GenAI through a Research Agent that generates contextual travel recommendations based on organisational policies, traveller preferences and business context. Workflow execution, approvals and state transitions are currently managed through automated deterministic orchestration to ensure transparency, auditability and enterprise governance. In future commercial releases with GDS integrations, Samanvita will incorporate Agentic AI through specialised Execution Agents capable of autonomously coordinating bookings, itinerary modifications, exception handling and enterprise system integrations, enabling end-to-end intelligent travel orchestration.

4.9 Customer Service Strategy

The customer service strategy is on similar lines of ISTM and is as follows:

Priority	Type	Issue Type	Target Response	Target Resolution / Service Restoration	Support Owner
P1	Incident	Complete platform outage, cloud infrastructure failure, database failure, major security incident or disaster recovery event affecting all enterprise customers	Within 15–30 minutes (24×7)	Immediate restoration with a target of ≤ 4 hours	Engineering, DevOps & Customer Success
P2	Incident	Booking execution failures, incorrect AI recommendations affecting travel decisions, integration failures or major performance degradation	Within 1 hour	Within 4–8 hours	Engineering & Product Support
P3	Incident	Approval workflow failures , Individual user issues, booking modifications, configuration problems, portal defects or access issues	Within 1 business day	Within 2–3 business days	Customer Success

P4	Service Request	Feature requests, user onboarding, access provisioning, training requests, documentation enquiries and other non-incident requests	Within 2 business days	As per agreed Service Request SLA	Customer Success & Product Team
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5. Marketing Plan

5.1 Demand Forecasting

The demand forecast is illustrated in the below table:

Category	Key Figures & Years	Sources	Summary
Market Trends	2023: Travel budgets increased across firms 2024: +18.3% growth; ~USD 38.2B 2025: USD 37.2B (2024); +15.5% projected	Financial Express FCM travel GBTA & VISA Study	Strong post-pandemic recovery with steady growth. Rising budgets and double-digit expansion indicate a stable and growing corporate travel market.
Market Demand	50,000+ daily AI interactions	MMT AI Assistant AI for Travel	Travel planning remains time-intensive, while AI adoption is rising. This shows demand for integrated, automated travel management solutions like Samanvita.
Revenue Potential	TAM: \$37.2B SAM: ~ \$1.86B (5% of TAM) SOM: ~ \$9.3M (0.5% of SAM) Y1 Revenue Estimate: \$2.3M Y2 Revenue Estimate: \$5.1M Y3 Revenue Estimate: \$9.3M	GBTA & VISA Study	Targets a focused segment of corporate travel. Revenue is projected using a bottom-up approach, aiming to capture ~0.5% of the serviceable accessible market by Year 3.

The market trend can also be illustrated using a line graph as follows:

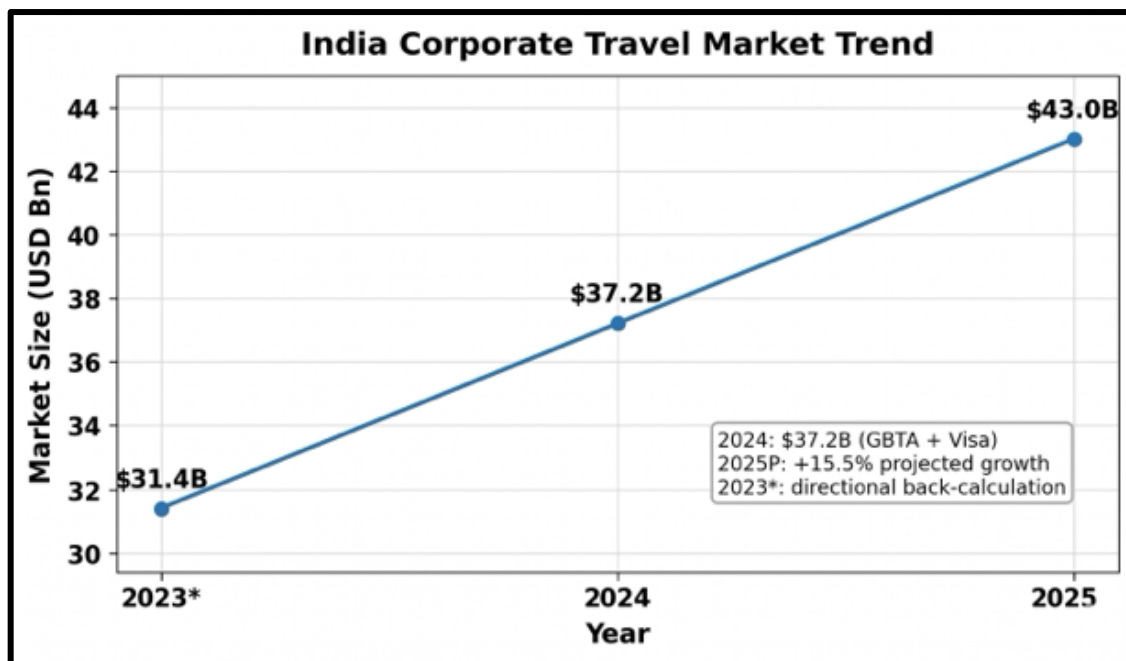


Fig 5.1.1: Demand Forecasting

5.2 Segmentation, Targeting and Positioning

Segmentation - The market was segmented using organisational characteristics, travel frequency, operational complexity and current travel management practices.

Segment	Organisation Profile	Travel Characteristics	Key Pain Points	Fit for Samanvita
High-frequency Enterprise Coordinators	Medium and large enterprises across IT Services, Consulting, Engineering, Manufacturing and Corporate Services	Weekly or multiple monthly business travel managed through email, travel agencies and booking platforms	Fragmented coordination, repeated approvals, booking delays, itinerary changes and low workflow visibility	High
Operations & Financial Approvers	Organisations with structured approval and finance processes	Periodic business travel requiring policy compliance and budget governance	Approval bottlenecks, expense handling, reporting and reconciliation	High
Founders & Lean Businesses	Small organisations with limited administrative resources	Low to moderate travel frequency with simple approval workflows	Limited visibility, manual coordination and administrative overhead	Medium

Large Global Enterprises	Large multinational organisations with mature enterprise systems	High-volume, multi-country corporate travel	Complex policies, ERP/HRMS/GDS integrations, security and compliance requirements	Future Target
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Targeting - Organisations with recurring employee business travel that still rely on multiple disconnected systems such as email, travel agencies, online booking platforms, ERP systems and messaging applications to coordinate travel approvals, bookings and itinerary management.

Target Users or Ideal Customer Profile - Travel Desk, Operations, Administration, HR, Finance

Positioning - Samanvita delivers seamless coalescence through one-click deployment by unifying travel requests, contextual AI recommendations, approvals, workflow execution and enterprise governance into a single connected platform. It transforms fragmented corporate travel processes into a streamlined, transparent and auditable end-to-end orchestration experience while keeping humans in control of critical business decisions.

5.3 Seven Ms

The 7Ms for Samanvita are as follows:

M	Samanvita Application
Market	Organisations with recurring employee business travel and fragmented travel coordination, approval and execution workflows.
Mission	Reduce the effort and delay between employee travel submission and confirmed execution while retaining enterprise governance.
Message	Seamless Coalescence through One-Click Deployment —Samanvita connects travel requests, contextual recommendations, approvals and workflow execution into one connected enterprise platform.
Media	Enterprise demos, LinkedIn thought leadership, webinars, case studies, partner referrals and industry events.
Money	Onboarding, subscription, seat, volume and complexity pricing; marketing investment prioritised toward pilots and enterprise sales.
Measurement	Qualified leads, pilot conversion, implementation time, CAC, ARR, NRR, Orchestration Success Rate and approval turnaround time.
Management	Product-led governance involving Product, Engineering, Customer Success, Security, Sales and enterprise implementation owners.

5.4 Channel Strategy

Samanvita will adopt a **direct enterprise go-to-market strategy** that combines digital lead generation with relationship-driven enterprise sales. The objective is to educate organisations about intelligent travel orchestration, demonstrate business value through product walkthroughs, and convert qualified prospects into pilot customers. As customer adoption grows, implementation support, customer success and case studies will drive account expansion and referrals.

Channel	Objective	Primary Audience
Company Website	Product information, lead generation and demo requests	Enterprise decision makers
LinkedIn	Thought leadership and outbound prospecting	HR, Operations, Travel Desk, Product Leaders
Webinars	Product education and live demonstrations	Prospective enterprise customers
Enterprise Demonstrations	Showcase workflow automation and AI capabilities	Decision makers and stakeholders
Industry Conferences & Events	Brand awareness, networking and lead generation	Corporate travel and HR professionals
Customer Referrals	Build credibility and acquire qualified leads	Existing enterprise customers

5.5 Mix of Online/Offline and Owned vs Franchised

Samanvita will operate through a **hybrid online-offline model** using **fully owned channels**. Customer acquisition and engagement will be managed directly by the company to ensure consistent product positioning, implementation quality and long-term customer relationships. At the current stage, franchising or reseller partnerships are not part of the business model.

Dimension	Approach
Online	Website, LinkedIn, webinars, email campaigns, virtual product demonstrations
Offline	Enterprise meetings, travel & HR conferences, networking events and product showcases
Owned Channels	Primary distribution strategy
Strategic Partnerships	Travel technology, enterprise software and implementation partners (future)

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5.6 Purely Online vs Omnichannel

Samanvita will adopt an **omnichannel strategy** because enterprise software purchasing involves multiple stakeholders and decision stages. Digital channels generate awareness and qualified leads, while offline interactions provide opportunities to demonstrate the product, understand organisational requirements and build long-term business relationships. This combination supports a more effective enterprise sales process than relying solely on online channels.

Stage	Primary Channel
Awareness	LinkedIn, Website, Industry Content
Consideration	Webinars, Product Videos, Case Studies
Evaluation	Enterprise Demonstrations, Meetings
Purchase	Sales Consultation & Pilot Implementation
Adoption	Customer Success & Onboarding
Expansion	Business Reviews & Referrals

5.7 Parity between Channels

Samanvita will maintain a consistent customer experience across all channels through standardised brand messaging, product demonstrations, implementation methodologies and customer success practices. Regardless of whether a customer engages online or offline, they will receive consistent product positioning, a tailored commercial proposal, a standardised onboarding process and a consistent support experience supported by shared knowledge resources and documented issue management processes.

Area	Approach
Brand Messaging	Standardised value proposition and product positioning across all channels
Product Demonstrations	Consistent demonstration environment and business workflows
Pricing	Customer proposals based on a common enterprise pricing methodology while accommodating organisation-specific requirements
Customer Onboarding	Standardised implementation and onboarding methodology
Support	Consistent customer success process, issue management and knowledge resources
Marketing Content	Unified sales collateral, case studies and product documentation

5.8 Go-To-Market (GTM) Strategy

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Samanvita's GTM strategy will be used for Product Launch. The initial GTM motion is founder-led and consultative because the product changes an operational workflow and requires buyer trust. Prospects should enter through a demonstration based on a realistic corporate travel scenario, followed by a short process-discovery workshop. The sales conversation should quantify approval delays, coordination hours, traveller volumes and exception rates before recommending a pilot. A pilot should begin with one department, travel purpose or traveller group and use agreed success metrics. Expansion should occur only after the organisation observes stable execution and measurable reduction in administrative effort.

5.9 Product Launch Strategy

Samanvita will follow a phased commercialisation strategy to minimise implementation risk and incorporate customer feedback before broader market adoption. The product will progress through internal validation, pilot deployments, a controlled commercial launch and enterprise scale-up, with each stage advancing only after predefined success criteria are achieved. This approach ensures that product reliability, onboarding capability and customer support mature alongside market expansion

Stage	Primary Objective	Key Activities	Success Criteria
Internal Testing	Validate product reliability	End-to-end workflow testing, bug fixes and technical validation	Core workflows operate reliably
Pilot Deployment	Validate business value	Deploy with selected organisations, gather feedback and refine workflows	Pilot objectives achieved and customer feedback incorporated
Controlled Commercial Launch	Begin customer acquisition	Onboard qualified enterprise customers and monitor implementation	Stable onboarding process and successful pilot outcomes
Enterprise Scale-up	Expand market presence	Grow customer base, strengthen support and enhance platform capabilities	Product, support and implementation processes become repeatable

5.10 Twelve-Month GTM and Product Launch Timeline

Samanvita's GTM strategy will be used for Product Launch. The initial GTM motion is founder-led and consultative because the product changes an operational workflow and requires buyer trust. Prospects should enter through a demonstration based on a realistic corporate travel scenario, followed by a short process-discovery workshop. The sales conversation should quantify approval delays, coordination hours, traveller volumes and exception rates before recommending a pilot. A pilot should begin with one department, travel purpose or traveller group and use agreed success metrics. Expansion should occur only after the organisation observes stable execution and measurable reduction in administrative effort.

Month	Product / Readiness	Marketing & Sales	Customer Milestone	Decision Gate
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1	Freeze MVP; prepare demo, security gap list and pilot checklist	Finalise website, one-pager, demo video and ICP list	20–30 qualified target accounts	MVP and demo stable
2	Prepare cloud architecture and pilot onboarding	Founder outreach, LinkedIn and warm introductions	8–10 discovery meetings	Validate customer pain and buyer ownership
3	Deliver authentication, RBAC and logging	Tailored demos and workflow workshops	2–3 pilot partners	Pilot success metrics agreed
4	Launch controlled pilot with assisted support	Educational content and workflow webinar	First live pilot	Validate workflow reliability
5	Refine onboarding, dashboards and alerts	Publish first case study and ROI narrative	First successful pilot completed	Demonstrate measurable operational improvement
6	Release Pilot V1 and integration roadmap	Soft commercial launch	First paid or extended pilot	Validate willingness to pay
7	Begin GDS and payment integration discovery	Account-based outreach and strategic partnerships	3–5 qualified opportunities	Prioritise integrations from customer demand
8	Enhance reporting, tenant configuration and customer success tools	Webinars, referrals and case study promotion	Second and third paying customers	Validate repeatable onboarding
9	Strengthen security, VAPT readiness and procurement documentation	Target larger enterprise accounts	First enterprise procurement review	Assess enterprise sales readiness
10	Prototype disruption and reminder workflows	Industry conferences and executive roundtables	Expansion within an existing customer	Validate land-and-expand strategy
11	Improve analytics, SLA monitoring and integration framework	Renewal planning and quantified case studies	Renewal intent from pilot customers	Validate long-term product value
12	Finalise Year-2 roadmap	Review CAC, conversion, NRR and channel performance	Repeatable sales and onboarding playbook	Scale based on retention and unit economics

5.11 Growth Hacks and Growth Loops

Growth Hacks - Samanvita will focus on low-cost, high-impact customer acquisition through founder-led sales, targeted LinkedIn outreach, industry conferences, webinars and pilot deployments. Early

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customer success stories and referrals will be leveraged to build credibility, generate qualified leads and accelerate enterprise adoption.

Growth Hack	Objective
Founder-led LinkedIn thought leadership and product demonstrations	Build credibility and generate qualified enterprise leads.
Industry conference participation and live workflow demonstrations	Increase brand awareness and acquire pilot customers.
Case studies and ROI calculators	Demonstrate measurable business value and improve conversion.
Referral incentives for pilot customers	Encourage customer advocacy and lower acquisition costs.
Educational webinars on AI-driven travel orchestration	Position Samanvita as a category expert and generate inbound demand.

Growth Loops - Samanvita will focus on improving customer outcomes, expanding platform adoption and strengthening long-term customer relationships through continuous product enhancement and proactive customer success. This approach will **increase Customer Lifetime Value (CLV) and renewal rates** by continuously delivering measurable operational value throughout the customer lifecycle.

Initiative	Expected Outcome
Continuous product improvements based on customer feedback	Increases customer satisfaction and long-term retention
Regular business reviews with enterprise customers	Demonstrates value delivered and identifies expansion opportunities
Continuous improvement of AI recommendation quality	Builds user trust and increases platform adoption
Adoption of additional workflows, integrations and enterprise modules	Expands platform usage and increases Customer Lifetime Value (CLV)
Proactive customer success, onboarding and support	Improves renewal rates and reduces customer churn

6. Finance Plan

6.1 Cost Structure

Revenue estimate of Y1 is \$2.3M. A buffer of nearly \$400k+ is assumed and the total operating expenses is set at ~\$1.8M. Thus, some of the key elements of the cost structure (**on a yearly basis**) can be tabulated as:

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S.No	Key Element of Cost Structure	% of Revenue (Benchmark)	Est. Annual Cost (USD)	Summary	Source
1	Product Development & Engineering	22%	\$506,000	Building the core workflow/execution engine and state managers.	SaaS Capital 2025
2	Sales, Onboarding & Enterprise Integration	21%	\$483,000	13% Selling / 8% Marketing: Enterprise sales and onboarding.	
3	Compliance, GST, and Security Costs	14%	\$322,000	G&A: Legal, DPDP, HR, and tax-management infra.	
4	Cloud, Infrastructure, and AI Costs	9%	\$207,000	5% Hosting / 4% DevOps: AWS and AI tokens.	
5	Operations, Support & Failure Handling	8%	\$184,000	Customer Success: 24/7 support and manual interventions.	
6	Travel API and Payment Integration	7%	\$161,000	5% Pro Services / 2% Other COGS: Travel APIs and Payment Gateways.	Razor Pay
Total		81%	\$1,863,000	Contingency Buffer: \$437,000 (19%)	

6.1 Revenue Model

The revenue model is – Hybrid Bottom Up model with subscription, seat-based, and usage-based components.

The breakdown is as follows:

- 1) Platform fee
- 2) Onboarding fee
- 3) Subscription fee –
 - a. Yearly subscription
 - b. Quarterly subscription
 - c. Monthly subscription
- 4) Per admin seat fee
- 5) Volume Based Usage Tier –
 - a. Standard – 1 to 10 Trips
 - b. Intermediate – 11 to 30 trips
 - c. Advanced – 31 to 60 trips
 - d. Premium – 61 to 100 Trips
 - e. Elite – Greater than 100 Trips

6) Complexity Tier –

Destinations \ Travellers	1–10 Travellers	11–30 Travellers	31–60 Travellers	>60 Travellers
1 City	Standard	Intermediate	Advanced	Professional
2–5 Cities	Intermediate	Advanced	Professional	Premium
6–10 Cities	Advanced	Professional	Premium	Elite
>10 Cities	Professional	Premium	Elite	Elite

7) Total Trip Cost = Number of Trips * Complexity Tier Cost

8) Additional Add Ons – Variable Cost

Total Revenue = Platform Fee + Onboarding Fee + Subscription Fee + (Per Admin Seat Cost × Number of Admin Seats) + (Number of Trips × Complexity Tier Cost) + Add-ons Cost

7. Enterprise Readiness: Reliability, Security & Compliance

The MVP architecture is suitable for demonstration and pilot learning but requires production hardening before enterprise deployment. Google Sheets should be replaced with a transactional database such as PostgreSQL, and ngrok should be replaced with stable cloud infrastructure. Authentication, organisation-level access control, secret management and workflow resilience must be strengthened to support commercial enterprise deployments.

Area	Production Readiness
Infrastructure	Replace Google Sheets with PostgreSQL and ngrok with cloud-native networking, monitoring and high availability.
Security	Enterprise SSO (SAML/OIDC), RBAC, encryption, managed secrets and immutable audit logs.
Reliability	Retry logic, timeout handling, idempotency controls, automated backups and disaster recovery.
Compliance	DPDP Act, GDPR readiness, ISO/IEC 27001, SOC 2 Type II, OWASP Top 10 and periodic VAPT.
Risk Management	Provider-independent integrations, configurable business rules, prompt versioning and alternate notification channels.

8. Ethical Considerations

Samanvita is designed around the principle that AI should augment enterprise decision-making rather than replace human judgement. Ethical AI usage requires transparency, fairness, privacy and accountability while ensuring that travel recommendations remain aligned with organisational policies and employee well-being.

Ethical Consideration	Samanvita Approach
Data Privacy	Process employee travel information only for authorised business purposes, with encryption, RBAC and applicable data protection compliance.
Transparency & Explainability	Every AI recommendation includes the rationale behind the suggested itinerary or decision.
Algorithmic Bias	Continuously evaluate AI outputs to minimise bias arising from training data and ensure consistent recommendations across employees.
Traveller Safety	Prioritise traveller well-being by considering factors such as women traveller safety, child travellers where applicable, secure accommodation, trusted transport providers and organisational travel policies.
Human Oversight	AI recommendations remain advisory, while approvals and critical business decisions are made by authorised personnel.
Responsible Data Usage	Employee information is processed with organisational authorisation, retained only as necessary and used solely to deliver enterprise travel services.

9. Future AI Roadmap

As the platform matures, Samanvita will progressively evolve from AI-assisted recommendations to specialised agentic capabilities that automate increasingly complex enterprise travel workflows under human oversight. Each agent will address a distinct business function while maintaining explainability, governance and enterprise control.

Version	AI Capability	Primary Function
V1 (Current MVP)	AI Research Agent	Contextual travel recommendations and itinerary generation.
V2	AI Compliance Agent	Interpret travel policies and validate policy compliance.
V3	AI Disruption Agent	Manage delays, cancellations and itinerary re-planning.
V4	AI Execution Agent	Coordinate bookings, modifications and enterprise system integrations.
V5	AI Expense & Reconciliation Agent	Automate expense validation, reconciliation and financial workflows.

V6	AI Budget & Forecasting Agent	Predict travel demand, optimise budgets and recommend cost-saving opportunities.
V7	Executive Travel Intelligence Agent	Deliver enterprise-wide insights, forecasting and strategic travel analytics.

10. Project Management

Product Management can be successfully executed using Project Management. They must work in tandem with each other.

10.1 Project Management Methodology

The project followed an **Agile** approach using the **Scrum** framework because requirements evolved iteratively through continuous development, testing and validation. Work was organised into three sprints, enabling infrastructure, backend orchestration and frontend integration to be delivered incrementally while accommodating changes identified during implementation.

10.2 Project Plan

The project plan implemented for the MVP is as follows:

Sprint	Duration	Objective	Key Activities	Deliverables
Sprint 1 – Project Setup & Infrastructure	Week 1	Establish the technical foundation and development environment	Set up Docker, n8n, Jira, Confluence, GitHub and Figma; configure Gemini API; create the Google Sheets datastore; install Postman; design the system architecture	Development environment established, project workspace configured and technical architecture finalised
Sprint 2 – Backend Orchestration & AI Workflow	Weeks 2–3	Develop the core backend orchestration workflow	Build the travel request intake workflow; implement the AI Research Agent; configure Google Sheets integration; develop approval workflows, email notifications, webhook communication and workflow state tracking	Functional backend orchestration with AI-powered recommendations and approval workflow

Sprint 3 – Frontend, Integration & Validation	Week 4	Complete the MVP through frontend integration, testing and documentation	Design the employee portal and dashboards; integrate the frontend with backend workflows; implement email action pages; perform end-to-end testing; prepare the PRD, capture screenshots and record the demonstration video	Fully functional MVP, validated workflows, project documentation and demonstration video
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10.3 JIRA Boards

The Jira Board screenshots are available at the this link - [JIRA Screenshots](#)

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